

Corollary 2.8 Extends to All Same-Generator Power Modules

Candidate full solution packet
arXiv:1610.01659, Question 2.10

11 August 2026

Status. Candidate full affirmative answer, subject to expert review. The result follows by applying the source's own representing- measure theorem and observing that the character positivity set of a power module depends only on its generators, not on the even exponent.

1 Source question

Let (A, τ) be a locally multiplicatively convex unital commutative real algebra. For a fixed family $G = \{g_j : j \in J\} \subset A$, define the $2d$ -power module

$$M_{2d}(G) := \sum A^{2d} + \left\{ \sum_{j \in F} g_j \sigma_j : F \subset J \text{ finite, } \sigma_j \in \sum A^{2d} \right\}. \quad (1)$$

Infusino and Kuhlmann ask in Question 2.10, printed on page 7 of [1], whether their Corollary 2.8 extends to the modules $M_{2d}(G)$ when the same nonempty generator set G is used for every d .

INFINITE DIMENSIONAL MOMENT PROBLEM: OPEN QUESTIONS AND APPLICATIONS. 7

Let us observe that once we have a positive answer to Question 2.6 (i.e. once we have identified some conditions weaker than continuity under which the non-negativity of L on some $2d$ -power module implies the existence of a representing measure for L), then the equivalence of (1), (2) and (3) can be established along the same lines of the proof of Corollary 2.8. However, these same methods do not provide an answer to the following question.

Question 2.10. *Can Corollary 2.8 be extended to $2d$ -power modules of A having the same non-empty set of generators?*

Theorem 2.4 can be viewed as a strengthening in the commutative case of the result in [43], Lemma 6.1 and Proposition 6.9 for enveloping algebras.

Explicitly, for a τ -continuous linear functional $L : A \rightarrow \mathbb{R}$, they ask whether the following are equivalent:

1. L is nonnegative on $M_2(G)$;
2. L is nonnegative on $M_{2d}(G)$ for every $d \in \mathbb{N}$;
3. L is nonnegative on $M_{2d}(G)$ for some $d \in \mathbb{N}$.

2 Proof intuition

The easy direction is purely algebraic: every $2d$ -th power is a square, so $M_{2d}(G) \subset M_2(G)$. For the reverse direction, suppose positivity is known for one exponent. Proposition 2.5 of the source turns continuity and $M_{2d}(G)$ -positivity into a nonnegative representing measure supported on the character positivity set of that module.

That support set is independent of d . Indeed, it is simply the set of characters at which every generator g_j is nonnegative. Every element of $M_2(G)$ is pointwise nonnegative there. Integrating therefore gives $M_2(G)$ -positivity. This is exactly the missing implication.

3 The exponent-independent positivity set

For a power module $M \subset A$, write

$$X_M := \{\alpha \in X(A) : \alpha(m) \geq 0 \text{ for all } m \in M\},$$

where $X(A)$ is the real character space of A .

Lemma 1. *For every $d \in \mathbb{N}$ and every generator family G ,*

$$X_{M_{2d}(G)} = K_G := \{\alpha \in X(A) : \alpha(g_j) \geq 0 \text{ for every } j \in J\}. \quad (2)$$

In particular, this set does not depend on d .

Proof. If $\alpha \in X_{M_{2d}(G)}$, then $g_j 1^{2d} \in M_{2d}(G)$, so $\alpha(g_j) \geq 0$ for every j . Conversely, suppose $\alpha \in K_G$. For every $a \in A$, $\alpha(a^{2d}) = \alpha(a)^{2d} \geq 0$, and

$$\alpha(g_j a^{2d}) = \alpha(g_j) \alpha(a)^{2d} \geq 0.$$

Linearity then makes α nonnegative on every finite sum in (1), proving $\alpha \in X_{M_{2d}(G)}$. $\square$

4 Main theorem

Theorem 2. *Let (A, τ) be an lmc unital commutative real algebra, let $G \subset A$ be any generator family (possibly infinite), and let $L : A \rightarrow \mathbb{R}$ be τ -continuous. Then conditions (1)–(3) listed above are equivalent. Hence Question 2.10 of [1] has an affirmative answer; in fact the proof also covers the empty generator family.*

Proof. For every d , the inclusion

$$\sum A^{2d} \subset \sum A^2$$

holds because $a^{2d} = (a^d)^2$. Multiplication by each common generator and taking finite sums preserves the inclusion, so $M_{2d}(G) \subset M_2(G)$. This proves (1) $\Rightarrow$ (2), while (2) $\Rightarrow$ (3) is immediate.

Assume (3), and fix d for which L is $M_{2d}(G)$ -positive. Proposition 2.5 of [1] gives a nonnegative representing measure μ for L with

$$\text{supp } \mu \subset X_{M_{2d}(G)} \cap \mathfrak{sp}(\tau) \subset K_G, \quad (3)$$

where the last equality is Lemma 1. Every $m \in M_2(G)$ has $\widehat{m}(\alpha) = \alpha(m) \geq 0$ on K_G , again by the lemma with $d = 1$. Therefore

$$L(m) = \int_{X(A)} \widehat{m}(\alpha) d\mu(\alpha) \geq 0 \quad (m \in M_2(G)).$$

Thus (3) $\Rightarrow$ (1), completing the equivalence. $\square$

5 Verification and novelty

The proof has only two substantive checks. First, the module convention in the source permits arbitrary generator families but only finite sums in each element; this is precisely (1), and Lemma 1 applies without any cardinality assumption. Second, source Proposition 2.5 applies to any $2d$ -power module of an lmc algebra, not only to the empty-generator case. The support inclusion it supplies is exactly (3), after which positivity on $M_2(G)$ is pointwise.

A bounded novelty search through 11 August 2026 covered the run indexes, the exact wording of Question 2.10, title and author searches, subsequent papers by the authors on infinite-dimensional moment problems, and combinations of “same generators,” “ $2d$ -power module,” and “Corollary 2.8.” No later paper explicitly answering Question 2.10 was found. The proof is a direct deduction from results printed immediately before the question; the novelty claim should therefore be treated cautiously even though no explicit answer was retrieved.

Human-review recommendation. Review the exact hypotheses of source Proposition 2.5 and the module definition. Under the printed statements, the argument is complete and needs no additional assumption such as finite generation, Archimedeaness, or completeness.

References

- [1] M. Infusino and S. Kuhlmann, *Infinite dimensional moment problem: open questions and applications*, Contemp. Math. **697** (2017), 187–201; arXiv:1610.01659.